ROBOTICS

How TRIC Robotics is reducing pesticide use on strawberries using UV light

Rebecca Szkutak — 7:00 AM PDT · July 23, 2025

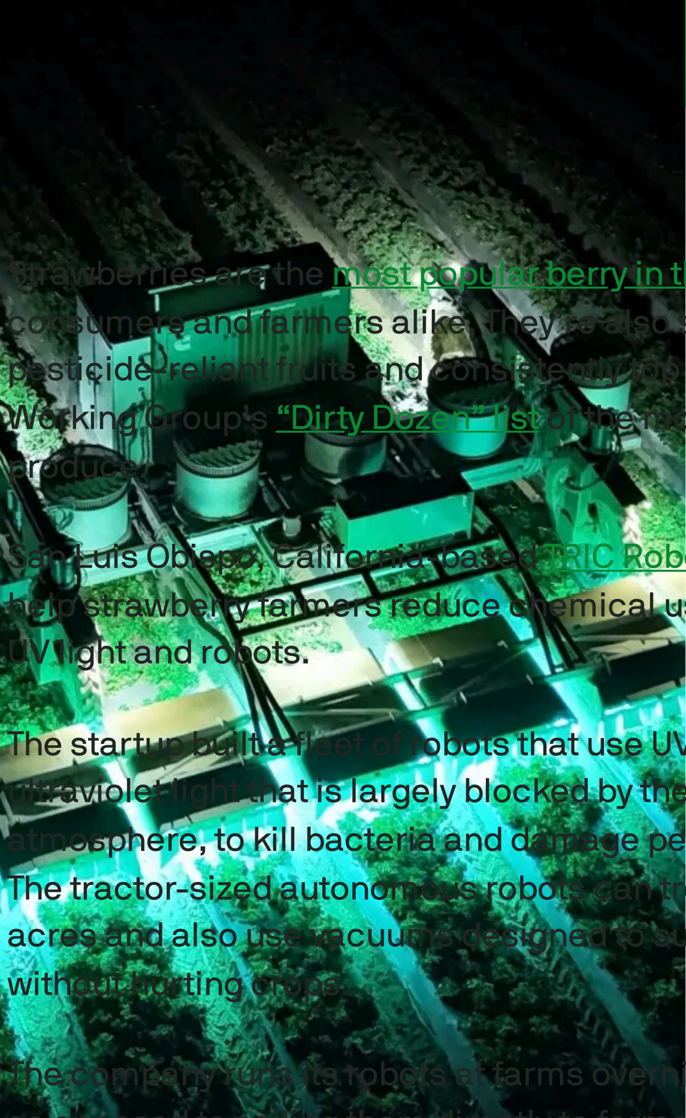

IMAGE CREDITS: TRIC ROBOTICS

Strawberries are the most popular berry in the U.S. for both consumers and farmers alike. They're also some of the most pesticide-reliant fruits and consistently top the Environmental Working Group's "Dirty Dozen" list of the most contaminated produce.

San Luis Obispo, California-based TRIC Robotics thinks it can help strawberry farmers reduce chemical use with the help of UV light and robots.

The startup built a fleet of robots that use UV-C light, a form of ultraviolet light that is largely blocked by the earth's atmosphere, to kill bacteria and damage pest populations. The tractor-sized autonomous robots can treat up to 100 acres and also use vacuums designed to suck up bug residue without hurting crops.

The company runs its robots at farms overnight as a service, as opposed to selling them directly to farmers, because, while harder to scale, this model seemed like the right one to start getting traction quickly, Adam Stager, the co-founder and CEO of TRIC, told TechCrunch.

"We worked a lot with the farmers to understand the right way to launch the technology and what was the right business model," Stager said. "We found out that a lot of the farmers pay for pest disease control as a service, so they have a company come in and do the sprays. And what we've been doing is just replacing that as a service model."

While Stager said the company has been very focused on what farmers want, it wasn't always that way. In fact, TRIC wasn't even focused on agriculture to begin with.

Stager launched the company in 2017 after completing his PhD in robotics. The company was initially focused on 3D-printed robots for SWAT teams. In 2020, Stager decided to pivot into an area he thought would have more impact and started focusing on agriculture.

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"I really just wanted to answer the question, if you were to die tomorrow, would you be happy with what you accomplished in your life?" Stager said. "I was like, okay, I really need to do something impactful that can help a lot of people to feel value for myself. I kind of stumbled into agriculture on that journey, [and realized] that's a place where we can impact so many people, just about everybody."

Stager reached out to the United States Department of Agriculture (USDA) to see if there was any technology they were working on that he could help commercialize, knowing from his PhD program that a lot of great technology never leaves the lab.

He got connected to a USDA program that brings folks like Stager and scientists, who haven't yet commercialized their work, together. This outreach connected him to the UV light technology that became the basis for TRIC's robotics.

"We loaded two robots that we built in my garage on top of the SUV," Stager said about him and co-founder Vishnu Somasundaram. "We had two connections that the USDA helped us build with farmers that were willing to give us just a tiny little piece of land in 2021 and that's really the beginning of when this company started. It was a cross-country journey of Airbnb surfing for eight months where we were deploying two robots and getting this amazing data with these farmers."

Now the company, which also counts Ryan Berard as its third co-founder, works with four large strawberry producers, has deployed nine robots, and has three more robots on the way.

TRIC Robotics recently raised a $5.5 million seed round led by Version One Ventures with participation from Garage Capital, Todd and Rahul Capital, and Lucas Venture Group, among other investment firms and individual angels.

The company plans to put the money toward continuing to build out its fleet of autonomous robots, and TRIC eventually wants to move into other types of crops as well.

“I think there is going to be a really, really bright future for [agriculture] tech,” Stager said. “I just think people should know that things are really headed in a great direction, and there’s really a lot of exciting things to come.”

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STARTUPS

Seso is building software to fix farm workforces and solve agriculture's HR woes

Rebecca Szkutak — 5:05 AM PDT · April 2, 2024

IMAGE CREDITS: DAVID S. HOLLOWAY / CONTRIBUTOR / GETTY IMAGES

Migrant workers are a critical labor force for U.S. farms, but getting them here on proper H-2A visas can be complicated, and the compliance surrounding these employees is taxing for farms. Seso was founded five years ago to help streamline that process and now looks to expand into a one-stop-shop HR platform for the agriculture industry.

Michael Guirguis co-founded the startup after his cousin asked for his advice on whether her organic farm should expand. Despite demand for her harvests, Guirguis, whose entire career has involved job creation and the labor market, told her expanding wouldn't be smart because the industry's labor shortage would make hiring enough workers hard. That inspired Guirguis to found Seso to automate the H-2A visa process to help fix that issue and help farms stay compliant. Once he started talking to potential farm customers, he realized that farms could use a lot more help with their HR beyond just finding workers.

While agriculture is a massive industry ripe for disruption, it's been relatively reticent to adopt new technology, he said. Guirguis thinks Seso has been successful in selling to farms so far, when many other startups haven't been, because Seso isn't trying to change the actual farming process, something farmers made clear to him that they weren't ready for yet. Adopting back office tech is an easier sell.

"Your HR team is in the back office doing traditional HR work," Guirguis said. "That is who we are trying to change behavior for, which is easier than for someone 50 years in the field still using pen and paper. They can still keep doing their process we have built products to adapt. You can take a picture of a [handwritten] time sheet and then use AI to make sure that is accurate."

Guirguis's focus on getting feedback from farmers directly is what pushed Nina Achadjian, a partner at Index Ventures, to invest. Achadjian initially passed on Seso when it first tried to raise from Index, but how the company sells and interacts with farmers changed her mind.

"I remember this one customer call, I got chills," Achadjian told TechCrunch. "[He said], 'I get pitched by these Silicon Valley entrepreneurs all the time and they show up at your farm and they are like, 'Here is how you should run their business.' I always ask each of them to come and spend a day and work alongside me so they can understand what is a day in the life of the end customer and they never show up. Michael was the only one who showed up at 4 a.m. in the freezing cold, in the dark, to pick artichokes.'"

That feedback from farmers is why the company is expanding into automating payroll next. Guirguis said due to various agriculture employment laws, farm payroll is incredibly complicated. Workers are paid for how much crop they pick, Guirguis said, and the rate for each crop picked is different for a migrant worker versus a domestic worker and different again if migrant workers and domestic workers are picking from the same field. Guirguis sees numerous ways to expand after that.

Topics: AgTech Fundraising North America SaaS Seso Startups Startups Venture venture capital

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Abundant Robotics spins out of SRI to bring apple-picking robots to the farm

Lora Kolodny — 7:00 AM PDT · August 10, 2016

IMAGE CREDITS: ONDROM / SHUTTERSTOCK

Apples are the second most commonly consumed fruit in the U.S., according to data from the U.S. Department of Agriculture, but the way apples are harvested in the orchard hasn't changed much in the past two centuries. It's still a manual process, by and large.

Now, SRI Ventures, a kind of startup incubator within the Menlo Park research and development firm SRI International, is investing in and spinning out a startup called Abundant Robotics Inc. to automate apple harvesting with agricultural robots currently in development.

The robots, as yet unnamed, were designed to be strong and fast enough to remove one fruit per second from a tree, but gentle enough not to damage trees or the fruit, according to Abundant Robotics' CEO and co-founder Dan Steere.

They employ computer vision to recognize apples on the branch that are ready for harvesting, and a kind of vacuum to remove the apples, Steere said.

For power, the robots are plugged into the small tractors already used pervasively in fruit farming.

Steere co-founded Abundant Robotics about a year ago, he says, with CTO Curt Salisbury and Senior Software Architect Michael Eriksen recognizing a massive market opportunity in agriculture.

The team had previously worked on research funded by SRI and additional, non-dilutive grants from the Washington Tree Fruit commission to bring automation to the field.

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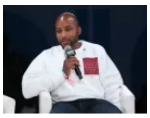

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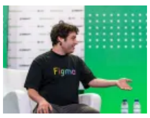

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Farmers typically produce more than 9 billion pounds of apples domestically, exporting 2.3 billion pounds of them in 2014-2015, according to the USDA. And international consumption of them has been growing.

Steere said, "Seeing fruit and picking it without damaging it is the big engineering challenge. If you bruise or cut the fruit it loses its value."

According to SRI Ventures President, Manish Kothari, it had not been possible to automate the task of apple picking before recent breakthroughs in computer vision and image processing were made.

He said, "You direct this robot to go someplace, see and pick an apple, and go again. It's a very non-trivial engineering challenge. To detect apples very precisely you have to see down at the millimeter level in real time. That requires software, and on the hardware side, chips that allow you to do real time image processing on the fly."

SRI Ventures in 2016 has already spun out 7 companies this year so far, he noted, including Superflex and others all of which have attracted outside, venture investments.

Abundant Robotics declined to disclose how much funding it had raised, but Kothari and Steere confirmed the startup has closed an equity seed round in addition to its grant funding.

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While most of the intellectual property the company developed so far was created at SRI, some was developed in partnership with robotics labs at Carnegie Mellon University, Steere said, and the company has a tech transfer agreement with the school.

Following its spin out from SRI Ventures, Abundant Robotics will aim to have its autonomous, robotic apple pickers in production and at work in orchards within 2 years, and grow its 8-person team currently based at SRI in Menlo Park, Calif.

The company has conducted field trials with orchards in Washington state, and overseas in Australia. It plans to engage in more of these prior to commercialization of its robots, Steere said.

Although a representative for Abundant Robotics claimed the startup wasn't releasing any images of its vacuum picker, a trade news site called Good Fruit Grower has one from a closed demo in March, here.

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